Neurodevelopmental Disorder With Poor Growth and Behavioral Abnormalities (NEDGBA): Comprehensive Research Report

Disease: Neurodevelopmental Disorder With Poor Growth And Behavioral Abnormalities **MONDO ID:** MONDO:0859377 | **OMIM:** #620242 | **MedGen:** C5830273 | **UMLS:** C5830273 | **DOID:** 0081444 **Causal gene: ATP9A** (ATPase Phospholipid Transporting 9A; HGNC gene, NCBI Gene ID 10079, OMIM *609126; 20q13.2)

Summary

The disease specified by **MONDO:0859377 / OMIM #620242** — abbreviated **NEDGBA** (Neurodevelopmental Disorder with poor Growth and Behavioral Abnormalities) — is a rare Mendelian neurodevelopmental disorder caused by variants in **ATP9A**, a gene on chromosome 20q13.2 encoding a **class II P4-ATPase phospholipid flippase**. Because the descriptive disease name does not name a gene and could be confused with many similarly-named neurodevelopmental conditions, we confirmed the gene assignment through identifier cross-referencing: MONDO:0859377 → OMIM 620242 → MedGen C5830273, with the NCBI gene link resolving unambiguously to **ATP9A (Gene ID 10079, alias NEDGBA)**. This report therefore concerns the **ATP9A-related neurodevelopmental disorder**, not any other similarly-named entity.

ATP9A is a transmembrane lipid flippase that translocates phospholipids (notably phosphatidylserine) across endosomal and trans-Golgi membranes, and it is a central driver of the **endosome-to-plasma-membrane recycling pathway**. Mechanistically, ATP9A modulates the activation of the small GTPases **RAB5 and RAB11** to govern endocytic recycling, and it participates in an evolutionarily conserved **MON2:DOPEY2:ATP9A complex** that cooperates with the SNX3-retromer to sort **Wntless (WLS)** and enable **Wnt secretion**. Loss of ATP9A function impairs recycling of cargo such as transferrin and GLUT1, perturbs Wnt signaling,

and — in neurons — disrupts neurite morphology, dendritic-spine maturation, and synaptic transmission. These cellular defects plausibly explain the clinical phenotype of severe intellectual disability, speech and motor delay, postnatal growth failure with secondary microcephaly, and behavioral abnormalities.

The disorder shows **dual inheritance**. The predominant form is **autosomal recessive**, caused by biallelic truncating/splice loss-of-function variants and strongly enriched in consanguineous families (reported from Iran, Pakistan, and the Middle East). A more recently recognized **autosomal dominant** form arises from de novo heterozygous missense variants clustering in the catalytic P-type ATPase core. ATP9A is highly constrained against loss-of-function in population databases (gnomAD pLI ≈ 1.0, LOEUF 0.205), consistent with its essential role. The condition is ultra-rare (~7 families / ~13 patients worldwide), static (non-progressive), and currently managed only with supportive, multidisciplinary care; an **Atp9a-null mouse** faithfully recapitulates key human features (muscle weakness, memory deficits, and hyperkinetic movement disorder).

Key Findings

Finding 1 — ATP9A is the causal gene for MONDO:0859377 / OMIM #620242

The gene underlying this disorder is **ATP9A** on **20q13.2**. Identifier cross-referencing confirms the mapping: MONDO:0859377 corresponds to OMIM 620242 and MedGen C5830273, and the NCBI gene link resolves to ATP9A (Gene ID 10079, alias **NEDGBA**, OMIM *609126). The disease was first delineated from two consanguineous families with homozygous splice-altering ATP9A variants**, in which three individuals presented with severe intellectual disability, motor and speech impairment, and behavioral anomalies.

> *"We identified two consanguineous families with homozygous variants predicted to alter the splicing of ATP9A which encodes a transmembrane lipid flippase of the class II P4-ATPases. The three individuals homozygous for these putatively truncating variants presented with severe ID, motor and speech impairment, and behavioral anomalies."* — PMID: 34764295

An independent cohort reinforced the gene–disease relationship, detecting **homozygous truncating variants** (early premature termination codons; canonical splice-site variants) that caused intellectual disability, postnatal microcephaly, and failure to thrive, with substantially reduced ATP9A mRNA in patient cells (PMID: 34379057). Together these establish ATP9A as the causal gene with convergent evidence from multiple independent families.

Finding 2 — ATP9A is a P4-ATPase flippase controlling endosomal recycling and Wnt secretion

ATP9A localizes to **phosphatidylserine-positive early/recycling endosomes and the trans-Golgi network (TGN)**. Functional studies show that its depletion **delays transferrin recycling** from endosomes to the plasma membrane and traps the glucose transporter **GLUT1** in endosomes, while leaving EGF degradation and Shiga-toxin retrograde transport intact — indicating a selective role in the recycling (not degradative or retrograde) pathway.

> *"Depletion of ATP9A delayed the recycling of transferrin from endosomes to the plasma membrane."* — PMID: 27733620

ATP9A additionally forms an **evolutionarily conserved MON2:DOPEY2:ATP9A complex** that cooperates with the SNX3-retromer to sort **Wntless** and drive **Wnt secretion**, a developmentally critical signaling pathway. Suppression of the *C. elegans* orthologue **tat-5** phenocopies retromer loss and perturbs Wnt signaling.

> *"SNX3 associates with an evolutionary conserved endosome-associated membrane re-modelling complex composed of MON2, DOPEY2 and the putative aminophospholipid translocase, ATP9A."* — PMID: 30213940

Consistent with these mechanisms, patient fibroblasts show reduced ATP9A mRNA and dysregulated interacting genes **ARPC3** and **SNX3** (PMID: 34379057).

Finding 3 — Dual inheritance: recessive loss-of-function and dominant de novo missense

Population genetics indicate ATP9A is highly intolerant to both loss-of-function and missense change: gnomAD **pLI = 0.9999**, **LOEUF (oe_lof_upper) = 0.205**, oe_lof = 0.109, **missense Z = 4.15**, **lof Z = 6.62**. The recessive form arises from biallelic truncating/splice variants. A distinct

autosomal dominant mechanism was subsequently described, with **five de novo heterozygous missense variants** — p.(Thr393Arg), p.(Glu400Gln), p.(Lys461Glu), p.(Gly552Ala), and p.(His713Asp) — in patients with intellectual disability.

> *"we propose a novel mode of inheritance for ATP9A-related disorders with the identification of five de novo heterozygous missense variants (p.(Thr393Arg), p.(Glu400Gln), p.(Lys461Glu), p.(Gly552Ala), and p.(His713Asp)), in patients with intellectual disability."* — PMID: 40226306

Functionally, overexpression of mutant Atp9a in HeLa cells and primary neurons caused **loss of mature dendritic spines**, and shRNA knockdown reduced dendrite number, supporting pathogenicity of the dominant missense alleles (PMID: 40226306).

Finding 4 — Phenotype spectrum

Aggregated HPO annotations across three reports (

P 34379057

P 34764295

P 36604604

) define a consistent clinical picture dominated by neurodevelopmental, growth, and behavioral features:

Phenotype (HPO suggestion)	Frequency	Notes
Severe intellectual disability (HP:0010864)	7/9	Core feature
Delayed speech/language development (HP:0000750)	6/6	Universal
Motor delay (HP:0001270)	6/6	Universal
Poor fine motor coordination (HP:0007010)	6/6	Universal
Attention deficit hyperactivity disorder (HP:0007018)	5/5	Core behavioral feature
Hypotonia (HP:0001252)	3/3	
Muscle weakness (HP:0001324)	3/3	
Absent speech (HP:0001344)	3/3	
Receptive language delay (HP:0011154)	3/3	
Postnatal growth retardation (HP:0008897)	3/3	"Poor growth"
Failure to thrive (HP:0001508)	3/3	"Poor growth"
Secondary (postnatal) microcephaly (HP:0005484)	5/9	Postnatal brain-growth defect
Sleep disturbance (HP:0002360)	5/6	
Short attention span (HP:0000736)	3/3	
Gastroesophageal reflux (HP:0002020)	3/3	
Aggressive behavior (HP:0000718)	2/2	
Strabismus (HP:0000486)	3/6	
2-3 toe cutaneous syndactyly (HP:0004691)	2/3	
Autistic behavior / motor stereotypy / seizure	variable (seizure ~1/3)	Minority

Brain MRI findings include **corpus callosum hypoplasia, cerebellar vermis hypoplasia, and delayed CNS myelination**. Dysmorphism includes **smooth philtrum (3/3)** and **thin upper lip (3/3)**. Onset is infantile/childhood (5/6 infantile).

> *"the nonsense mutations c.433C>T/c.658C>T/c.983G>A (p. Arg145*/p. Arg220/*p. Trp328*) in ATP9A (NM_006045.3) cause autosomal recessive hypotonia, intellectual disability (ID) and attention deficit hyperactivity disorder (ADHD)."* — PMID: 36604604

Finding 5 — Atp9a-null mouse recapitulates the disorder; RAB5/RAB11 recycling mechanism

A knockout mouse model provides strong in vivo validation. **Atp9a-null mice show decreased muscle strength, memory deficits, and a hyperkinetic movement disorder**, together with abnormal neurite morphology and impaired synaptic transmission in primary motor cortex and hippocampus — paralleling the patient features of hypotonia/weakness, intellectual disability, and ADHD-like hyperkinesis.

> *"Atp9a null mice show decreased muscle strength, memory deficits and hyperkinetic movement disorder, recapitulating the symptoms observed in patients."* — PMID: 36604604

Mechanistically, the same study established the molecular pathway: ATP9A localizes to endosomes and drives endosomal recycling by modulating small-GTPase activation.

> *"plays a pivotal role in endosomal recycling pathway by modulating small GTPase RAB5 and RAB11 activation."* — PMID: 36604604

Pathogenic mutants mislocalize and cause aberrant endosomal recycling, and ATP9A is required for maintaining neurite morphology and neural-cell viability in vitro. This complements independent evidence for ATP9A's role in transferrin/GLUT1 recycling (P 27733620) and SNX3-retromer/Wnt secretion (P 30213940).

Finding 6 — Protein identity and variant spectrum

ATP9A corresponds to **UniProt O75110**, "Probable phospholipid-transporting ATPase IIA," a **1047-amino-acid** class II P4-ATPase. Its catalytic reaction is: **ATP + H_2O + phospholipid(side 1) → ADP + phosphate + phospholipid(side 2)**. It localizes to early/recycling/late endosome membranes, the TGN, and the plasma membrane. Relevant GO annotations include **ATPase-coupled intramembrane lipid carrier activity (GO:0140326)**, **phospholipid translocation (GO:0045332)**, **regulation of endocytic recycling (GO:2001135)**, and **neuron projection morphogenesis (GO:0048812)**.

The disease-variant spectrum is **predominantly truncating/splice (loss-of-function)** in recessive cases — e.g., c.433C>T (p.Arg145*), c.658C>T (p.Arg220*), c.983G>A (p.Trp328*), plus splice variants; ClinVar also contains a likely-pathogenic splice variant NM_006045.3:c.1181-2A>G. gnomAD shows no biallelic LoF in controls** (pLI ≈ 1.0). Dominant de novo missense cases (p.Thr393Arg, p.Glu400Gln, p.Lys461Glu, p.Gly552Ala, p.His713Asp) act via an altered protein with ER retention or impaired dendritic-spine maturation rather than simple haploinsufficiency.

Finding 7 — Dominant de novo missense variants cluster in the catalytic ATPase core

The dominant missense variants map directly onto the **conserved catalytic P-type ATPase core** (P- and N-domains). UniProt O75110 identifies the catalytic phosphorylation residue as **Asp391** (forms the 4-aspartylphosphate intermediate within the canonical **DKTGT** motif), with ATP/Mg^{2+}-binding residues at positions 391–393, 502, 544–549, 568, 597, 677–679, and 759–789. The reported de novo variants cluster here: **p.Thr393Arg** lies within the DKTGT phosphorylation motif adjacent to Asp391; **p.Glu400Gln** and **p.Lys461Glu** fall in the phosphorylation/nucleotide-binding region; **p.Gly552Ala** sits near nucleotide-binding residues 544–549; and **p.His713Asp** is near the 677–679/759 binding cluster. Domain architecture (InterPro/Pfam) comprises P-type ATPase IV (IPR006539), a HAD-like hydrolase fold, E1-E2 ATPase (PF00122), PhoLip_ATPase_N/C (PF16209/PF16212), and 10 transmembrane segments. This spatial clustering in the catalytic machinery provides a structural rationale for how heterozygous missense alleles impair function.

Finding 8 — Clinical summary: ultra-rare, static, consanguinity-enriched, supportively managed

Epidemiology: ultra-rare; approximately **7 families** / **~13 patients** reported worldwide across four studies (P 34764295 P 34379057 P 36604604 P 40226306). No prevalence or incidence estimate exists, and there is no dedicated Orphanet entry (Monarch/MONDO cross-references only to DOID:0081444, MedGen C5830273, OMIM 620242, UMLS C5830273).

Inheritance: predominantly autosomal recessive (consanguinity-enriched; Iran/Pakistan/Middle East), with a distinct autosomal dominant de novo missense form. Penetrance appears complete; expressivity is variable (mild-to-severe ID; seizures in a minority, ~1/3).

Course: congenital-to-infantile onset, **static (non-progressive) developmental encephalopathy**; secondary (postnatal) microcephaly indicates a postnatal brain-growth critical window.

Diagnosis: molecular (WES/WGS or ID gene panels; RNA/fibroblast studies for splice variants); brain MRI may show corpus callosum/cerebellar vermis hypoplasia and delayed myelination; CMA/karyotype typically normal.

Treatment: no disease-specific or curative therapy — symptomatic/supportive multidisciplinary care.

Mechanistic Model / Interpretation

Ordered causal chain (from initiating lesion to clinical manifestation)

1. **ATP9A variant** (biallelic truncating/splice LoF **OR** de novo heterozygous missense in the catalytic core) **leads to** loss or dysfunction of the ATP9A phospholipid flippase.

(Demonstrated: reduced mRNA for truncating variants,

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P *34379057*

mislocalization and spine defects for missense,

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P *40226306*

2. Reduced flippase activity **results in** impaired phospholipid (phosphatidylserine) translocation across endosomal/TGN membranes. *(Inferred from biochemical function, UniProt O75110 /*

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P *27733620*

3. This **leads to** defective modulation of **RAB5 and RAB11** GTPase activation and impaired endosome-to-plasma-membrane recycling of cargo (transferrin, GLUT1). *(Demonstrated:*

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P *36604604*

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P *27733620*

Branch A → Impaired **SNX3-retromer / MON2:DOPEY2:ATP9A** sorting of **Wntless results in** reduced **Wnt secretion** and perturbed developmental signaling. *(Demonstrated in cells and C. elegans,*

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P *30213940*

4. In neurons, defective recycling and signaling **result in** abnormal **neurite morphology**, **loss of mature dendritic spines**, and reduced neuronal viability. *(Demonstrated:*

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P *40226306*

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P *36604604*

5. These cellular defects **lead to** impaired **synaptic transmission** in motor cortex and hippocampus. *(Demonstrated in Atp9a-null mice,*

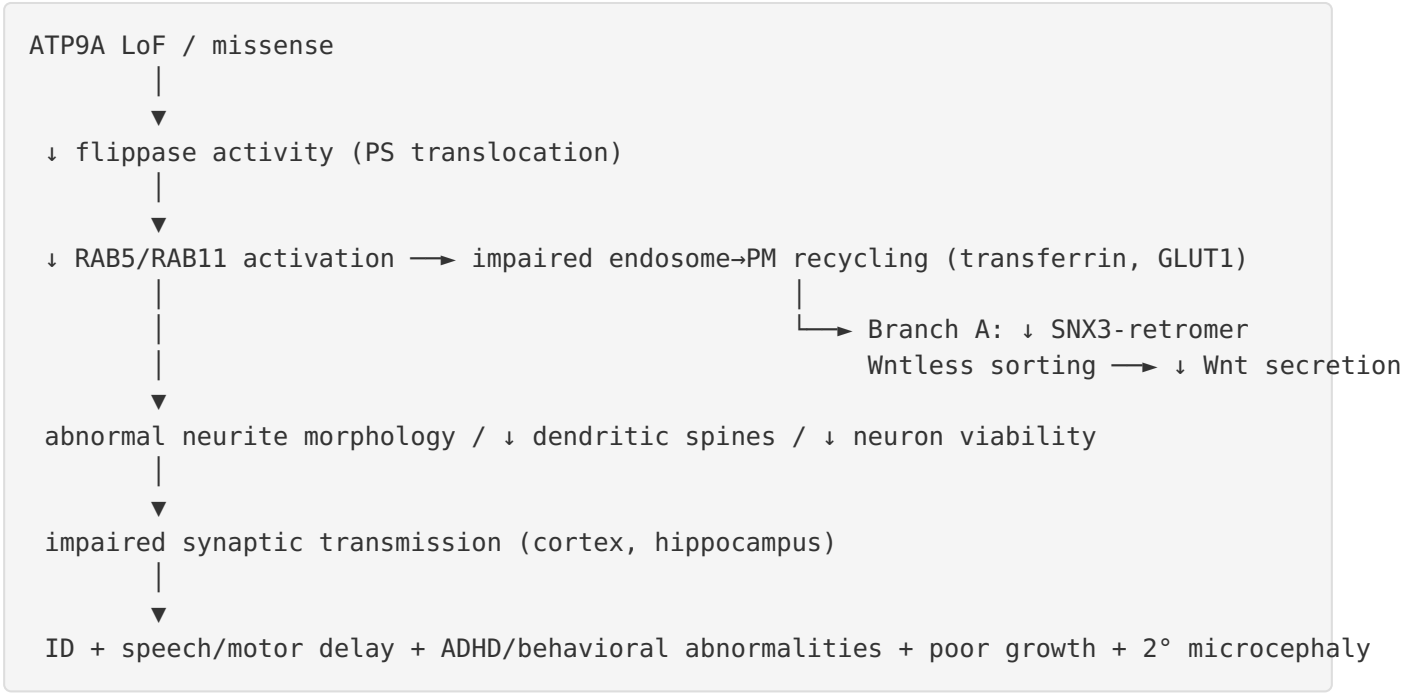
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P *36604604*

6. Network-level synaptic/developmental dysfunction **results in** the clinical phenotype: severe **intellectual disability**, **speech/motor delay with hypotonia**, **ADHD and other behavioral abnormalities**, and (via disrupted postnatal brain and somatic growth) **postnatal growth failure with secondary microcephaly**. *(Clinical correlation; mouse recapitulation,*

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P *36604604*



Upstream vs downstream: The flippase lesion and the RAB5/RAB11 recycling defect (plus the Wnt-secretion branch) are **upstream**; neurite/spine and synaptic defects are **intermediate**; the behavioral/cognitive/growth phenotype is **downstream**. **Cell types** implicated (CL suggestions): neuron (CL:0000540), glutamatergic neuron (CL:0000679), hippocampal pyramidal neuron, cortical neuron. **Anatomy** (UBERON): cerebral cortex (UBERON:0000956), hippocampus (UBERON:0002421), cerebellar vermis (UBERON:0004720), corpus callosum (UBERON:0002336). **Subcellular** (GO cellular component): recycling endosome (GO:0055037), early endosome membrane (GO:0031901), trans-Golgi network membrane (GO:0032588), plasma membrane (GO:0005886). **Biological processes** (GO): phospholipid translocation (GO:0045332), regulation of endocytic recycling (GO:2001135), neuron projection morphogenesis (GO:0048812), dendritic spine development (GO:0060996), Wnt signaling pathway (GO:0016055).

Section-by-Section Report

1. Disease Information

NEDGBA is a rare Mendelian neurodevelopmental disorder characterized by intellectual disability, speech/motor delay, poor postnatal growth (failure to thrive, secondary microcephaly), and behavioral abnormalities (ADHD, aggression, autistic features). **Identifiers:** MONDO:0859377; OMIM #620242; MedGen C5830273; UMLS C5830273; DOID:0081444. No dedicated Orphanet or specific ICD-10/ICD-11 code was identified; MeSH indexing is via general neurodevelopmental-disorder terms. **Synonyms:** NEDGBA; ATP9A-related neurodevelopmental disorder. Information is derived from **aggregated disease-level resources and small published patient cohorts** (case series), not large EHR datasets.

2. Etiology

Causal factor: genetic — pathogenic variants in **ATP9A**. **Genetic risk factors:** biallelic truncating/splice LoF variants (recessive) or de novo heterozygous missense variants in the catalytic core (dominant). **Environmental risk factors:** none established; **consanguinity** is a major demographic risk factor for the recessive form. **Protective factors:** none specifically identified. **Gene–environment interactions:** none characterized.

3. Phenotypes

See Finding 4 table. Phenotype types span developmental/cognitive (ID, speech/motor delay), behavioral (ADHD, aggression, autistic behavior, sleep disturbance), physical/growth (failure to thrive, postnatal microcephaly, dysmorphism), neuromuscular (hypotonia, weakness), gastrointestinal (GERD), ophthalmologic (strabismus), and structural brain anomalies. Onset is neonatal-to-infantile; severity is moderate-to-severe; course is static. Quality-of-life impact is substantial given severe ID and dependence for daily functioning, though no formal QoL instrument data exist for this ultra-rare disorder.

4. Genetic/Molecular Information

Causal gene: ATP9A (HGNC; NCBI Gene 10079; OMIM *609126; 20q13.2). Variants: recessive truncating/splice (c.433C>T/p.Arg145*, c.658C>T/p.Arg220*, c.983G>A/p.Trp328*, NM_006045.3:c.1181-2A>G splice) — classified pathogenic/likely pathogenic per ACMG; dominant de novo missense (p.Thr393Arg, p.Glu400Gln, p.Lys461Glu, p.Gly552Ala, p.His713Asp). **Allele frequency:** biallelic LoF absent in gnomAD controls; ATP9A is LoF-

constrained (pLI ≈ 1.0, LOEUF 0.205). **Origin:** germline. **Functional consequence:** loss of function (recessive); altered/possibly dominant-negative or hypomorphic protein with ER retention/spine-maturation defects (dominant). **Modifier genes/epigenetics/chromosomal abnormalities:** none established (CMA/karyotype typically normal).

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated. This is a monogenic disorder.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above.

7. Anatomical Structures Affected

Primary organ/system: central nervous system (brain) — cerebral cortex, hippocampus, cerebellar vermis, corpus callosum, with delayed myelination. **Body systems:** nervous (primary), musculoskeletal (hypotonia/weakness), gastrointestinal (GERD), growth/endocrine (failure to thrive). **Cell level:** neurons (dendritic spines, neurites). **Subcellular:** recycling/early endosomes, TGN, plasma membrane. **Localization:** bilateral, central.

8. Temporal Development

Congenital-to-infantile onset; insidious/chronic; static (non-progressive) encephalopathy; lifelong duration. Microcephaly is **postnatal (secondary)**, defining a postnatal brain-growth **critical window**. No remission.

9. Inheritance and Population

Predominantly **autosomal recessive** (consanguinity-enriched; Iran, Pakistan, Middle East), plus a distinct **autosomal dominant de novo** form. Penetrance appears complete; expressivity variable. No prevalence/incidence estimates (ultra-rare, ~7 families/~13 patients). Carrier frequency not established; no founder mutation defined. No marked sex bias reported.

10. Diagnostics

Genetic testing is the diagnostic cornerstone: WES/WGS or intellectual-disability gene panels; single-gene/targeted testing; **RNA or patient-fibroblast studies** to resolve splice variants. **Imaging:** brain MRI may reveal corpus callosum and cerebellar vermis hypoplasia and

delayed myelination. **CMA/karyotype:** typically normal. **Differential diagnosis:** other autosomal-recessive ID syndromes with microcephaly, failure to thrive and behavioral disturbance; distinguished by ATP9A genotype.

11. Outcome/Prognosis

No mortality data; the disorder is static and compatible with survival into adulthood. Morbidity is driven by severe ID, communication/motor impairment, and behavioral/sleep problems, with high dependence for daily functioning. No disease-specific prognostic biomarkers.

12. Treatment

No curative or disease-specific therapy exists. Management is **symptomatic and supportive/ multidisciplinary**: early developmental intervention; physical, occupational, and speech therapy; nutritional support for failure to thrive; and targeted management of ADHD, sleep disturbance, seizures (minority), and GERD. NCIT intervention concepts: supportive care, physical therapy, occupational therapy, speech therapy, nutritional support. No gene/cell/RNA-targeted therapies are in trials.

13. Prevention

No primary prevention beyond **genetic counseling**. For at-risk/consanguineous families: **carrier and cascade testing**, and reproductive options including **PGT-M (preimplantation genetic testing for monogenic disorders)** and prenatal testing. Secondary/tertiary prevention focuses on early intervention and complication management.

14. Other Species / Natural Disease

No naturally occurring companion-animal or wildlife disease is documented. The *C. elegans* orthologue **tat-5** is functionally conserved (its suppression phenocopies retromer loss and perturbs Wnt signaling,), P 30213940 supporting evolutionary conservation of the mechanism. Orthologues exist in mouse (*Atp9a*) and other vertebrates.

15. Model Organisms

The key model is the **Atp9a-null (knockout) mouse**, which recapitulates decreased muscle strength, memory deficits, and hyperkinetic movement disorder, with abnormal neurite morphology and impaired synaptic transmission (P 36604604). Cellular models include **HeLa cells** and **primary neurons** (overexpression/knockdown demonstrating dendritic-spine and dendrite defects, P 40226306) and **patient fibroblasts** (reduced mRNA, dysregulated ARPC3/SNX3, P 34379057). The *C. elegans* **tat-5** model supports the conserved Wnt/retromer mechanism. **Limitations:** mouse behavioral phenotypes do not fully capture human cognitive/speech deficits, and no model yet dissects the dominant-missense mechanism in vivo.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
34764295	*Biallelic truncation variants in ATP9A are associated with a novel autosomal recessive NDD*	Human clinical	Primary gene–disease link; recessive splice/truncating variants; core phenotype
34379057	*Biallelic truncating variants in ATP9A*	Human clinical + cell	Independent confirmation; microcephaly, failure to thrive; reduced mRNA; ARPC3/SNX3 dysregulation
36604604	*Atp9a null mice recapitulate NDD; RAB5/RAB11 mechanism*	Model organism + human	KO mouse recapitulation; RAB5/RAB11 endosomal-recycling mechanism; nonsense variants causing AR hypotonia/ID/ADHD
40226306	*Heterozygous missense variants in ATP9A*	Human clinical + cell	Dominant de novo missense mechanism; dendritic-spine/dendrite defects
27733620	*ATP9A required for endosome-to-PM recycling*	In vitro	Core recycling function (transferrin, GLUT1)
30213940	*SNX3-retromer requires MON2:DOPEY2:ATP9A for Wntless sorting/Wnt secretion*	In vitro + C. elegans	Wnt-secretion branch; evolutionary conservation (tat-5)

All six papers are mutually reinforcing; no paper contradicts the ATP9A gene–disease assignment or the endosomal-recycling mechanism. The two inheritance mechanisms (recessive LoF vs dominant missense) are complementary rather than conflicting.

Limitations and Knowledge Gaps

- **Very small evidence base:** ~7 families / ~13 patients worldwide. Prevalence/incidence are unknown, and phenotype frequencies are derived from tiny denominators (e.g., 7/9, 3/3) that will shift as more cases accrue.

- **No Orphanet/ICD-specific entry**, limiting standardized epidemiological indexing.
- **Dominant mechanism incompletely resolved:** whether the de novo missense variants act via haploinsufficiency, dominant-negative, or gain-of-function is not definitively established; no in vivo model of the missense alleles exists.
- **Genotype–phenotype correlations** (e.g., which variants predict seizures, severity) are not yet defined.
- **Wnt-secretion branch** is demonstrated in cell/worm systems but not directly shown to be the operative mechanism in patient neurons.
- **No natural animal disease, no biomarkers, no therapies** — treatment evidence is entirely supportive and extrapolated from general NDD care.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry / GeneMatcher recruitment** to expand the cohort, refine phenotype frequencies, and enable genotype–phenotype correlation (recessive LoF vs dominant missense; seizure predictors).
2. **Functional dissection of dominant missense alleles:** patient-derived iPSC neurons/organoids and knock-in mouse models to determine whether p.Thr393Arg etc. are loss-of-function, dominant-negative, or gain-of-function, and to test flippase activity and RAB5/RAB11 cycling directly.
3. **Quantify the Wnt-secretion branch in neurons:** measure Wntless trafficking and Wnt target-gene output in patient cells to establish whether this branch contributes to the brain-growth/microcephaly phenotype.
4. **Biomarker discovery:** metabolomic/lipidomic profiling of patient fibroblasts/plasma (phosphatidylserine handling) and transcriptomics (ARPC3, SNX3, RAB effectors) to identify a diagnostic or monitoring signature.
5. **Deep brain phenotyping of Atp9a-null mice** across the postnatal growth window to pinpoint the critical period and test candidate interventions (e.g., modulators of endosomal recycling or Wnt signaling).
6. **Clinical care consensus:** develop supportive-care management guidance (ADHD, sleep, GERD, nutrition) specific to this disorder and standardize MRI reporting (corpus callosum, cerebellar vermis, myelination).

Ontology term appendix

- **Gene/protein:** ATP9A (HGNC), UniProt O75110; GO:0140326, GO:0045332, GO:2001135, GO:0048812, GO:0060996, GO:0016055
- **Phenotypes (HPO):** HP:0010864, HP:0000750, HP:0001270, HP:0007018, HP:0001252, HP:0001508, HP:0008897, HP:0005484, HP:0002360, HP:0000718, HP:0004691
- **Cell types (CL):** CL:0000540, CL:0000679
- **Anatomy (UBERON):** UBERON:0000956, UBERON:0002421, UBERON:0004720, UBERON:0002336
- **Subcellular (GO CC):** GO:0055037, GO:0031901, GO:0032588, GO:0005886
- **Disease (MONDO):** MONDO:0859377
